UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

General Certificate of Education O Level

MARK SCHEME for the June 2005 question paper

5090 BIOLOGY

5090/06 Paper 6 (Alternative to Practical), maximum mark 40

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UNIVERSITY of CAMBRIDGE
International Examinations

JUNE 2005

GCE O Level

MARK SCHEME

MAXIMUM MARK: 40

SYLLABUS/COMPONENT: 5090/06

BIOLOGY
Paper 6 (Alternative to Practical)

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1 (a) (i) 150 - 280; mm; (or equivalent cm.) **[2]**

(ii) image of that part of field of view for one blind spot;
falls on receptors of other eye;
blind spots do not coincide;
brain 'fills' the blank; A.W. **[up to 2]**
ignore ref. stereoscopic

(b) (i) iris correct;
pupil correct; (allow 1 if both correct on adjacent drawing) **[2]**

(ii) outer iris diameter same in both drawings;
pupil smaller in second drawing; **[2]**
R: sections

(iii) retina detects more light entering eye;
causing dazzle ref. somewhere;
ref. iris involvement
circular muscles (of iris) contract;
radial muscles relax; (allow 1 if both mentioned but confused)
smaller pupil;
ref. reflex/role of brain;
controls entry of light; **[up to 6]**
A: correct muscle function enlarging or constricting pupil

[Total: 14]

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2 (a) (i) blood; **[1]**

(ii)&(iii)together. Drawing marks: **[D.4]**

1 Cells at least 2 cm clear, clean, realistic
2 Two cells drawn in each correct space, round and oval
3 Central area shown pale **(ii)**
4 Nuclei realistic **(iii)**

Labels: In 2.1 - two from: **[1]**
cytoplasm; depression/pale area; plasma/cell/outer membrane; (blood) plasma;

In 2.2 - nucleus and
one of cytoplasm/cell/outer membrane; **[1]**

R: one of label marks if cell wall mentioned
[Up to 3 max of above marks if drawings transposed]

(b) For each of 2 contrasting pairs:
small(er) - large(er);
no nucleus - nucleus present;
pale stain/cytoplasm - darker;
human – non-human
10/more cells – 3/fewer cells;
round - oval; **[2]**
Two from: cytoplasm; cell membrane; rounded/regular;
both are animal cells; **[2]**

[Total: 11]

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3 **(a)** **(i)** lengths: 50.5; 47.5;
changes: +0.5; -2.5; **[4]**

(ii) Table 3.2.

1.0	S3
0.8	S2
0.6	S5
0.4	S1
0.2	S4

[All correct = 4]
[S2 below S3; S5 below S2; S1 below S5; S4 below S1]

(b) Two from: keep s/a of chips the same;
use more chips; use longer chips;
all from same potato/to eliminate variation;
equal/constant temperature;
repeat/replicate; use other soln. concs;
measure mass/weight;
repeat measures to constant values
cover to reduce evaporation (of solvent);
o.v.p. **[2]**

(c) **(i)** ***1** x axis labelled 'conc/mol dm^3'
***2** y axis labelled 'change in length/mm'
3 all plots clear and correct
4 curve well drawn and smooth **[4]**
* allow on bar graph

(ii) 0.35 (0.325 - 0.375) mol/dm^3; (max 3 decimal pts) **[1]**

[Total: 15]

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